# Data Science

## Lecture 09: Association Rule Mining

Prof. Dr. Michael Granitzer

# Where We Left Off

Lecture 08 grouped *objects* by similarity. Today groups *items* by co-occurrence — and every measure is a probability from Lecture 02 in disguise.

**Support**
a joint probability

→

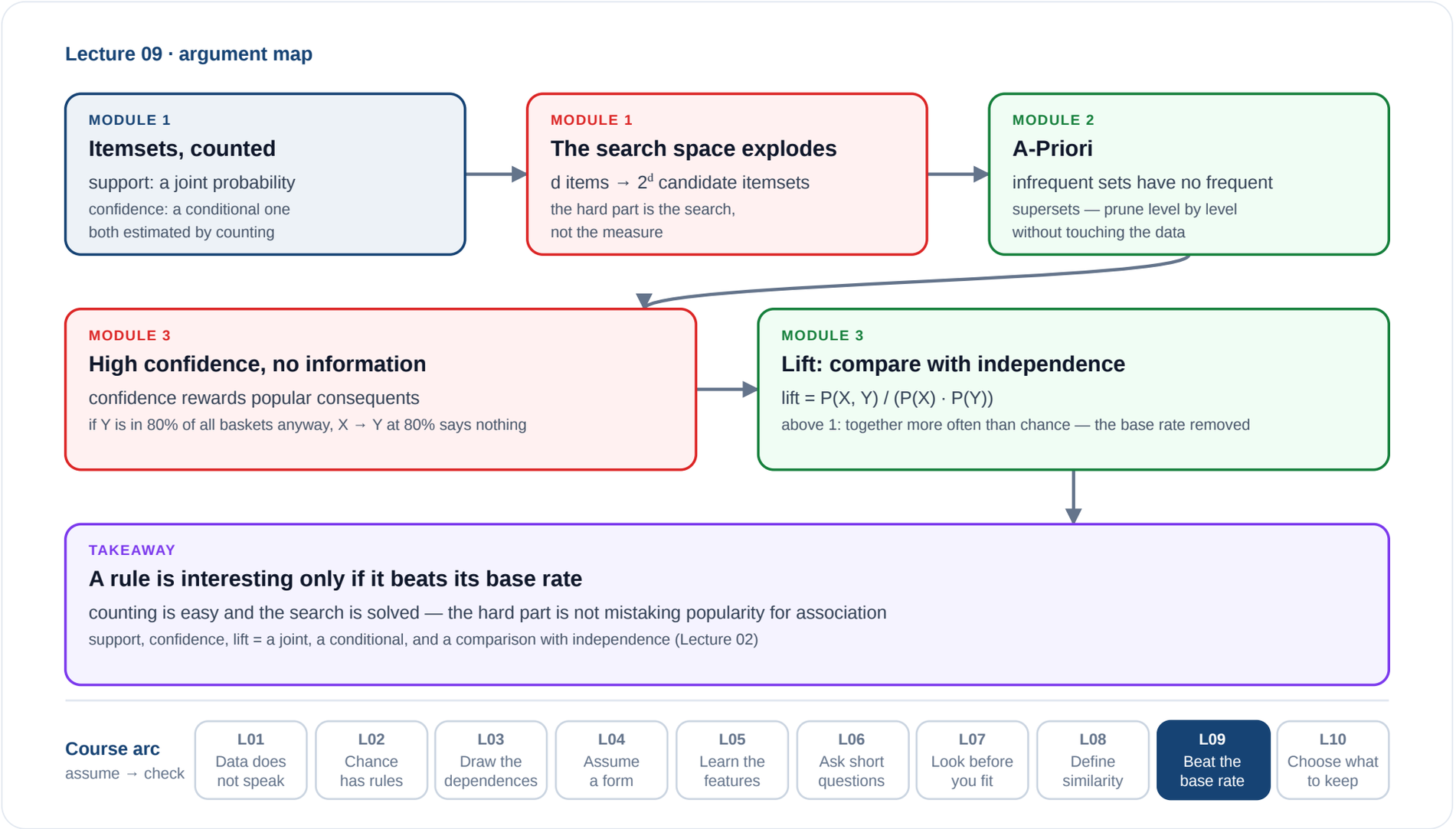
**Confidence**
a conditional one

→

**Lift**
a comparison with independence

> The base-rate trap from Lecture 02 returns: a rule can hold 80% of the time and say nothing.

# Argument Map



# Module 1

## Itemsets and Their Search Space

> ***Guiding Question:*** *Counting how often items occur together is easy. Why does it become hard with ten thousand products?*

# The Question

Given a set of transactions — shopping baskets, clicked pages, co-occurring symptoms — which **items appear together** more often than chance, and which **imply** which?

### Support

$$P(A \cap B) \geq \sigma_{\min}$$

How often do these items occur in the same transaction?

### Confidence

$$P(B \mid A) \geq \gamma$$

Given $A$ is in the transaction, how often is $B$ too?

Both are quantities the students already know: support is a **joint** probability, confidence a **conditional** one, both estimated by counting transactions.

# Formalisation

Let $I$ be a set of **items** and $X$ a set of **transactions**, each $x \subseteq I$.

- An **itemset** is any $A \subseteq I$; a $k$**-itemset** has $|A| = k$.
- Its **support** is $\sigma(A) = \frac{|\{x \in X : A \subseteq x\}|}{|X|}$.
- $A$ is **frequent** if $\sigma(A) \geq \sigma_{\min}$.

$\sigma(A \cup B)$ counts baskets that contain **all** items of $A$ **and** of $B$. A union of *itemsets* is an intersection of *events*: $\sigma(A \cup B) = P(A \cap B)$, not "A or B".

# The Search Space

All itemsets over $I$ form a **lattice** ordered by inclusion — with $2^{|I|}$ elements.

A supermarket with 10,000 products has $2^{10000}$ candidate itemsets. Enumeration is impossible; the entire chapter is about avoiding it.

Support and confidence are joint and conditional probabilities estimated by counting. The difficulty is not the measure but the **size of the search space**.

# Module 2

## The A-Priori Algorithm

> ***Guiding Question:*** *What single structural fact about support lets us discard most of an exponential space without looking at the data?*

# The Property That Makes It Possible

> **A-priori property (anti-monotonicity).** Every subset of a frequent itemset is frequent — equivalently: if an itemset is **infrequent, every superset of it is infrequent too**.

Adding an item can only ever *reduce* the number of transactions containing the set. So an infrequent itemset prunes its entire superset subtree from the search.

# The Algorithm

**A-Priori**

$L^1$ = all frequent 1-itemsets.

**for** $k = 2$; $L^{k-1} \neq \emptyset$; $k{+}{+}$: **generate** candidates $C_k = \{a \cup b \mid a \in L^{k-1}, b \in L^1, b \not\subseteq a\}$ **prune** any candidate having an infrequent subset **count** the support of the survivors in one pass over the transactions $L^k = \{c \in C_k \mid \sigma(c) \geq \sigma_{\min}\}$

### Generate

build $k$-candidates from frequent $(k-1)$-itemsets

### Prune

discard candidates with any infrequent subset — no data access needed

### Count

one database pass per level

# Cost and Improvements

### What it costs

One pass over the transactions **per level** $k$ , plus candidate generation. The bottleneck is I/O and the number of candidates.

### How it is improved

Hashing candidates into buckets, transaction reduction (dropping transactions containing no frequent itemset), partitioning and sampling.

A-Priori turns one structural fact — **infrequent sets have no frequent supersets** — into a level-wise search that prunes most of an exponential space without touching the data.

# Mini-Exercise

$\{\text{bread}, \text{milk}\}$ has support $0.18$, and $\sigma_{\min} = 0.22$.

**What do you know about $\{\text{bread}, \text{milk}, \text{eggs}\}$ — and how many database passes did it take to know it?**

# Mini-Exercise

$\{\text{bread}, \text{milk}\}$ has support $0.18$, and $\sigma_{\min} = 0.22$.

**What do you know about $\{\text{bread}, \text{milk}, \text{eggs}\}$ — and how many database passes did it take to know it?**

It is **infrequent**: adding eggs cannot raise the support above $0.18$. And it took **no pass at all** — the property alone decides. That is where A-Priori's savings come from.

# Module 3

## From Itemsets to Rules

> ***Guiding Question:*** *A rule holds 80% of the time — and tells you nothing. How is that possible, and what should you measure instead?*

# Generating Rules

An **association rule** $A \Rightarrow B$ (with $A \cap B = \emptyset$) has

$$\text{support} = \sigma(A \cup B), \qquad \text{confidence} = \frac{\sigma(A \cup B)}{\sigma(A)}$$

Every frequent itemset $F$ is split in all possible ways into $A$ and $B = F \setminus A$; rules meeting the confidence threshold are kept.

Support comes from the itemset and is already known. Only confidence must be computed — so rule generation needs **no further database passes**.

# Confidence Is Not Enough

Suppose 80% of all transactions contain milk. A rule $\text{bread} \Rightarrow \text{milk}$ with **confidence 0.80** has learned nothing: milk appears in 80% of baskets regardless of bread.

$$\text{lift}(A \Rightarrow B) = \frac{P(B \mid A)}{P(B)} = \frac{\sigma(A \cup B)}{\sigma(A)\sigma(B)}$$

Lift compares the joint probability against the product of the marginals — the **definition of independence** from Lecture 02, written as a ratio. It is symmetric: $\text{lift}(A \Rightarrow B) = \text{lift}(B \Rightarrow A)$.

# Reading Lift

| lift | meaning |
|---|---|
| $> 1$ | **positive association**: $A$ and $B$ co-occur more often than independence predicts |
| $= 1$ | **independent**: the rule carries no information |
| $< 1$ | **negative association**: $A$ and $B$ co-occur less often than independence predicts |

> Lift is a ratio, not a test. On a rule with low support it rests on a handful of baskets, and a lift of 1.5 can be chance. Before acting, check it for **significance** — the hypothesis tests of Lecture 02 apply.

# Mini-Exercise

A retailer reports $\mathrm{nappies} \Rightarrow \mathrm{beer}$ with support 0.02 and confidence 0.60; beer is in 50% of all baskets.

**What is the lift — and why is the rule still a weak basis for action?**

# Mini-Exercise

A retailer reports $\text{nappies} \Rightarrow \text{beer}$ with support 0.02 and confidence 0.60; beer is in 50% of all baskets.

## What is the lift — and why is the rule still a weak basis for action?

$\text{lift} = 0.60/0.50 = 1.2$: beer is 20% more likely with nappies. But the rule covers only 2% of baskets, so the effect may be chance — test it, and check that it persists.

# Mini-Exercise

A retailer reports $\text{nappies} \Rightarrow \text{beer}$ with support 0.02 and confidence 0.60; beer is in 50% of all baskets.

## What is the lift — and why is the rule still a weak basis for action?

$\text{lift} = 0.60/0.50 = 1.2$: beer is 20% more likely with nappies. But the rule covers only 2% of baskets, so the effect may be chance — test it, and check that it persists.

> Confidence alone rewards **popular** consequents. **Lift** corrects for the base rate by comparing against independence — the same correction Bayes' theorem makes for priors. Low support calls for a significance check.

# The Churn Shop's Baskets

The shop delivers grocery boxes. Of **100** boxes, **80** contain milk, **20** contain cereal, **16** contain both.

**The rule $\mathrm{cereal} \Rightarrow \mathrm{milk}$ holds with confidence 80%. Is it worth a "customers who order cereal also order milk" banner?**

# The Churn Shop's Baskets

The shop delivers grocery boxes. Of **100** boxes, **80** contain milk, **20** contain cereal, **16** contain both.

**The rule $\text{cereal} \Rightarrow \text{milk}$ holds with confidence 80%. Is it worth a "customers who order cereal also order milk" banner?**

$$\text{conf} = \frac{16}{20} = 0.8 \qquad \text{lift} = \frac{0.8}{0.8} = 1.0$$

No: milk is in 80% of **all** boxes. Cereal buyers take milk exactly as often as everyone else — the rule restates the base rate.